ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** retina	Light-sensitive screen at the back.
2	**(d)** a concave lens	The concave lens diverges the rays.
3	**(b)** dispersion	Newton's prism experiment.
4	**(d)** outer surface of the cornea	The cornea has the greatest change in refractive index.
5	**(a)** a convex lens	A convex lens converges the rays.
6	**(c)** Red	Red has the longest wavelength.
7	**(c)** iris	Iris adjusts the amount of light.
8	**(a)** in front of the retina	Eyeball is too long or lens too powerful.
9	**(b)** Violet	Shortest wavelength.
10	**(c)** accommodation	Done by ciliary muscles.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ability of the eye lens to adjust its focal length (1); To see both near and distant objects clearly (1).	1+1
12	Excessive curvature of the eye lens (1); Elongation of the eyeball (1).	1+1
13	Scattering of light by colloidal particles (1); Makes the path of light visible (1).	1+1
14	Iris (1); It changes the size of the pupil (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cornea: transparent front; most refraction (1); Iris and pupil control the light entering (1); Retina is the screen with rods and cones (1).	1+1+1
16	Ciliary muscles weaken (1); The eye lens loses flexibility (1); The near point recedes (presbyopia) (1).	1+1+1
17	They are much closer to us (1); They are extended sources made of many point sources (1); The variations from them average out (1).	1+1+1
	OR: For near objects they contract and the lens becomes thicker (1); Focal length decreases (1); For distant objects they relax and the lens becomes thinner (1).	1+1+1
18	Clouding of the lens of the eye (1); Causes loss of vision (1); Treated by surgery to replace the lens (1).	1+1+1
	OR: Light is deflected in many directions by particles (1); Blue colour of the sky (1); Red colour of the sun at sunrise (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cornea: transparent window that refracts light (1); Iris: controls pupil size (1); Pupil: allows light to enter (1); Lens: focuses the image on the retina (1); Retina and optic nerve: convert light into electrical signals sent to the brain (1).	1+1+1+1+1
	OR: Dispersion is splitting of white light (1); Light passes through a prism (1); Seven colour spectrum is formed (1); A second inverted prism recombines the colours (1); White light emerges, showing the prism does not create colours (1).	1+1+1+1+1
20	Accommodation is the adjustment of focal length (1); For near objects the lens becomes thicker (1); For far objects the lens becomes thinner (1); Near point 25 cm (1); Far point at infinity (1).	1+1+1+1+1
	OR: Sunlight enters a droplet and refracts (1); Dispersion splits it into colours (1); Internal reflection at the back of the droplet (1); Second refraction as it leaves (1); Observer sees red on top and violet at the bottom (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.

- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.
- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** farther than 25 cm	A person cannot see nearby objects clearly.
2	**(d)** red, orange, yellow, green, blue, indigo, violet	Red bends least and violet bends most.
3	**(a)** real and inverted	The brain interprets it as erect.
4	**(a)** weakening of ciliary muscles and loss of flexibility of the lens	It appears with age.
5	**(d)** dispersion, refraction and internal reflection by water droplets	Raindrops act as tiny prisms.
6	**(b)** 25 cm	The near point is 25 cm.
7	**(d)** bifocal lenses	Both concave and convex lenses.
8	**(c)** behind the observer	The observer sees colours from the opposite side.
9	**(b)** at infinity	Normal eye sees distant objects.
10	**(b)** cataract	Corrected by surgery.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hypermetropia (1); Focal length of the lens too long or the eyeball too short (1).	1+1
12	Starlight undergoes atmospheric refraction (1); Air density varies, so apparent position and brightness flicker (1).	1+1
13	Far point is at infinity (1); Near point is 25 cm (1).	1+1
14	Defect of vision with age (1); Near point recedes because ciliary muscles weaken (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Near-sightedness: the person cannot see far objects clearly (1); Causes: lens too curved or eyeball elongated (1); Corrected with a concave lens (1).	1+1+1
16	Splitting of white light into its component colours (1); Violet, indigo, blue, green, yellow, orange, red (1); Done by a prism (1).	1+1+1
17	Atmospheric molecules are smaller than the wavelength of visible light (1); They scatter shorter wavelengths more (1); Blue light reaches our eyes from all directions (1).	1+1+1
	OR: Far-sightedness: cannot see nearby objects clearly (1); Causes: focal length too long or eyeball too short (1); Corrected with a convex lens (1).	1+1+1
18	Refractive index of glass differs for different colours (1); Violet bends most, red least (1); So the colours emerge in different directions (1).	1+1+1
	OR: There is no atmosphere in space (1); So there is no scattering (1); The sky looks dark (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Near-sightedness (1); Image of a distant object is formed in front of the retina (1); Causes: excessive curvature of the lens, elongated eyeball (1); Corrected by a concave lens (1); The lens diverges rays so the image falls on the retina (1).	1+1+1+1+1
	OR: Air near the earth is optically denser at lower layers (1); Starlight bends towards the normal (1); Apparent position of the star changes and flickers (1); Sun is seen about 2 minutes before actual sunrise (1); Sun is seen about 2 minutes after actual sunset (1).	1+1+1+1+1
20	Far-sightedness (1); Image of a nearby object is formed behind the retina (1); Causes: long focal length or small eyeball (1); Corrected by a convex lens (1); The lens converges rays so the image falls on the retina (1).	1+1+1+1+1

	OR: Scattering depends on the size of the particles and wavelength (1); Blue light of short wavelength is scattered more (1); Sky appears blue (1); At sunrise light travels a long path in the atmosphere (1); Most blue light is scattered away so red reaches the eye (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.
- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** atmospheric refraction of starlight	Varying density of air.
2	**(d)** contract and the lens becomes thicker	Focal length decreases.
3	**(b)** concave lens of power -0.5 D	P = -1/2 m.
4	**(c)** they are nearer and act as extended sources	Variations average out.
5	**(d)** Retina	Rods and cones.
6	**(d)** eyeball too short	Focal length too long or eyeball short.
7	**(a)** atmospheric refraction	Rays bend towards the normal.
8	**(b)** distant objects clearly	Near-sightedness.
9	**(c)** -2 D	P = 1/f(m).
10	**(d)** 2 minutes	Due to atmospheric refraction.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water droplets are present in the air (1); Droplets disperse sunlight and return it to the observer (1).	1+1
12	To focus on objects at different distances (1); The image distance on the retina is fixed (1).	1+1
13	Blue light has short wavelength (1); It is scattered more by atmospheric molecules (1).	1+1
14	Light travels a long path through air (1); Blue is scattered away and red reaches us (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Myopia (1); f = -1.2 m (1); P = -0.83 D (concave lens) (1).	1+1+1
16	Water droplets act as small prisms (1); Sunlight is refracted, dispersed, internally reflected and refracted again (1); Different colours reach the observer from different droplets (1).	1+1+1
17	Scattering by colloidal particles in a medium (1); Path of light becomes visible (1); Light through a canopy of a dense forest or sun's rays through a smoky room (1).	1+1+1
	OR: Near point is within 25 cm or normal (1); The strong lens can focus close objects on the retina (1); The defect affects only distant vision (1).	1+1+1
18	Refraction of light by the earth's atmosphere (1); Twinkling of stars (1); Advance sunrise and delayed sunset (1).	1+1+1
	OR: Red has the longest wavelength (1); It is scattered least by fog or smoke (1); So it can be seen from a long distance (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	u = -25 cm, v = -100 cm (1); 1/f = 1/v - 1/u = -1/100 + 1/25 (1); 1/f = 3/100, f = 33.3 cm (1); P = +3 D (1); Convex lens (1).	1+1+1+1+1
	OR: (a) No atmosphere in space, so no scattering (1); (b) Red is scattered least (1); (b) Seen from far in fog or smoke (1); (c) Atmospheric refraction near the horizon (1); Rays from the lower and upper edges are refracted differently (1).	1+1+1+1+1
20	Image of a far object must be at 80 cm (1); u = infinity, v = -80 cm (1); 1/f = 1/v (1); f = -80 cm (1); P = -1.25 D (1).	1+1+1+1+1
	OR: Scattering of light by colloidal particles (1); The path of a beam becomes visible (1); Colour depends on the size of the scattering particles (1); Very fine particles scatter blue light (1); Larger particles scatter longer wavelengths, white at the largest (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.

- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
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Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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